

MTH100

Mid Semester Exam

Oct 9th, 2024

Time: 1 hour 20 minutes

Max Marks: 50

Instructions:

1. Attempt all the five questions. Marks for each question are indicated against it.
2. All your intermediate steps and calculations must be clearly shown.
3. You are not allowed to use determinant in this exam.
4. Marks for proof-type questions will depend on the logical progression of the steps. You may quote without proof any proposition or theorem covered in the lectures and tutorials but it must be clearly identified. Any other results used must be proved.
5. Start each question on a fresh side and clearly indicate if part of the question is done in a different part of the answer book.

Problem 1. (a) Are the vectors
 $\overline{\alpha}_1 = (1, 1, 2, 4), \overline{\alpha}_2 = (2, -1, -5, 2), \overline{\alpha}_3 = (1, -1, -4, 0) \text{ and } \overline{\alpha}_4 = (2, 1, 1, 6)$
 linearly independent in $\mathbb{R}^4$?

(5 points)

(b) **Show** that the following vectors in $\mathbb{R}^3$ form a basis for $\mathbb{R}^3$.
 $\overline{\beta}_1 = (1, 0, -1), \overline{\beta}_2 = (1, 2, 1) \text{ and } \overline{\beta}_3 = (0, -3, 2).$

 Express $\overline{e}_1 = (1, 0, 0)$ and $\overline{e}_3 = (0, 0, 1)$ as linear combinations of $\overline{\beta}_1, \overline{\beta}_2, \overline{\beta}_3$. (7 points)

Problem 2. (a) Using elementary row operations find the inverse of $A = \begin{bmatrix} 2 & 2 \\ 2 & 0 \end{bmatrix}$ (if it exists) over $\mathbb{Z}_3$ (4 points)
 (Note: You are not allowed to use determinant.)

(b) Extend $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \right\}$ to a basis for $\mathbb{R}^{2 \times 2}$ where $\mathbb{R}^{2 \times 2}$ is the vector space of all (2×2) matrices with real entries. (4 points)
 (Note: You need to show proper reasoning.)

Problem 3. (a) Let $C^1[-1, 1] = \{f \in C[-1, 1] : f'(x) \text{ exists and } f'(x) \text{ is continuous on } [-1, 1]\}$
 Is $C^1[-1, 1]$ a subspace of $C[-1, 1]$?
 If yes, is $C^1[-1, 1]$ a proper subspace of $C[-1, 1]$?
 (Give reasons) (4 points)

- (b) Let $R_3(t)$ be the vector space of all the polynomials (in variable t) of degree ≤ 3 with real coefficients. Let $f(t)$ be a polynomial of degree 3 in $R_3(t)$.

Show that for any $g(t) \in R_3(t)$, there exists scalars $c_0, c_1, c_2, c_3 \in \mathbb{R}$ such that

$$g(t) = c_0 f(t) + c_1 f'(t) + c_2 f''(t) + c_3 f'''(t) \quad (4 \text{ points})$$

- (c) (i) An $n \times n$ matrix A is called idempotent if $A^2 = A$.

Show that the only invertible idempotent $n \times n$ matrix is the identity matrix.

- (ii) Prove that if A and B are square matrices and AB is invertible, then both A and B are invertible.

(4 points)

Problem 4. (a) Let $\{v_1, v_2, \dots, v_n\}$ be a basis for a vector space V . Prove that $\{v_1, v_1 + v_2, v_1 + v_2 + v_3, \dots, v_1 + v_2 + \dots + v_n\}$ is also a basis for V (4 points)

- (b) Let M be an $n \times n$ upper triangular matrix with non-zero diagonal entries. Prove that the columns of M are linearly independent. (4 points)

Problem 5. Let A and B be $m \times n$ matrices that are both in RREF form such that $A \neq B$. Suppose that the first $(n - 1)$ columns of A and B are identical.

Assume further that neither A nor B have pivot positions in the last column.

Prove or disprove: there exists a vector $\bar{x}$ in $\mathbb{R}^n$ such that $A\bar{x} = \bar{0}$ but $B\bar{x} \neq \bar{0}$. (10 points)